- **Linear representations are the natural outputs of obvious algorithms a layer might implement.** If one sets up a neuron to pattern match a particular weight template, it will fire more as a stimulus matches the template better and less as it matches it less well.
- **Linear representations make features "linearly accessible."** A typical neural network layer is a linear function followed by a non-linearity. If a feature in the previous layer is represented linearly, a neuron in the next layer can "select it" and have it consistently excite or inhibit that neuron. If a feature were represented non-linearly, the model would not be able to do this in a single step.
- **Statistical Efficiency.** Representing features as different directions may allow *non-local generalization* in models with linear transformations (such as the weights of neural nets), increasing their statistical efficiency relative to models which can only locally generalize. This view is especially advocated in some of Bengio's writing (e.g. [5]). A more accessible argument can be found in this blog post.

It is possible to construct non-linear representations, and retrieve information from them, if you use multiple layers (although even these examples can be seen as linear representations with more exotic features). We provide an example in the appendix. However, our intuition is that non-linear representations are generally inefficient for neural networks.

One might think that a linear representation can only store as many features as it has dimensions, but it turns out this isn't the case! We'll see that the phenomenon we call *superposition* will allow models to store more features – potentially many more features – in linear representations.

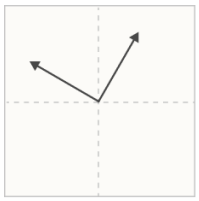
For discussion on how this view of features squares with a conception of features as being multidimensional manifolds, see the appendix “What about Multidimensional Features?”.

Privileged vs Non-privileged Bases

Even if features are encoded as directions, a natural question to ask is which directions? In some cases, it seems useful to consider the basis directions, but in others it doesn't. Why is this?

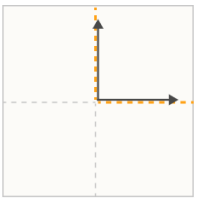
When researchers study word embeddings, it doesn't make sense to analyze basis directions. There would be no reason to expect a basis dimension to be different from any other possible direction. One way to see this is to imagine applying some random linear transformation M to the word embedding, and apply M^{-1} to the following weights. This would produce an identical model where the basis dimensions are totally different. This is what we mean by a *non-privileged basis*. Of course, it's possible to study activations without a privileged basis, you just need to identify interesting directions to study somehow, such as creating a gender direction in a word embedding by taking the difference vector between "man" and "woman".

But many neural network layers are not like this. Often, something about the architecture makes the basis directions special, such as applying an activation function. This "breaks the symmetry", making those directions special, and potentially encouraging features to align with the basis dimensions. We call this a privileged basis, and call the basis directions "neurons." Often, these neurons correspond to interpretable features.



In a **non-privileged basis**, features can be embedded in any direction. There is no reason to expect basis dimensions to be special.

Examples: word embeddings, transformer residual stream



In a **privileged basis**, there is an incentive for features to align with basis dimensions. This doesn’t necessarily mean they will.

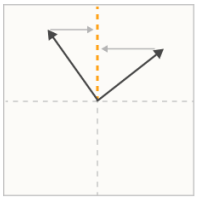
Examples: conv net neurons, transformer MLPs

From this perspective, it only makes sense to ask if a *neuron* is interpretable when it is in a privileged basis. In fact, we typically reserve the word "neuron" for basis directions which are in a privileged basis. (See longer discussion here.)

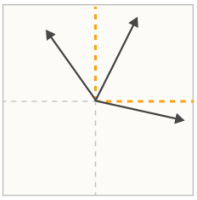
Note that having a privileged basis doesn't guarantee that features will be basis-aligned – we'll see that they often aren't! But it's a minimal condition for the question to even make sense.

The Superposition Hypothesis

Even when there is a privileged basis, it's often the case that neurons are "polysemantic", responding to several unrelated features. One explanation for this is the *superposition hypothesis*. Roughly, the idea of superposition is that neural networks "want to represent more features than they have neurons", so they exploit a property of high-dimensional spaces to simulate a model with many more neurons.



Polysemanticity is what we'd expect to observe if features were not aligned with a neuron, despite incentives to align with the privileged basis.

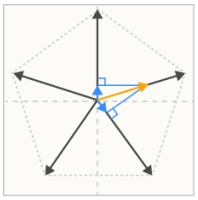


In the **superposition hypothesis**, features can't align with the basis because the model embeds more features than there are neurons. Polysemanticity is inevitable if this happens.

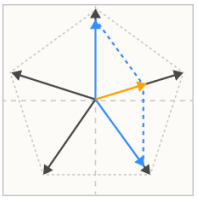
Several results from mathematics suggest that something like this might be plausible:

- **Almost Orthogonal Vectors.** Although it's only possible to have n orthogonal vectors in an n-dimensional space, it's possible to have $\exp(n)$ many "almost orthogonal" ($< \epsilon$ cosine similarity) vectors in high-dimensional spaces. See the Johnson–Lindenstrauss lemma.
- **Compressed sensing.** In general, if one projects a vector into a lower-dimensional space, one can't reconstruct the original vector. However, this changes if one knows that the original vector is sparse. In this case, it is often possible to recover the original vector.

Concretely, in the superposition hypothesis, features are represented as almost-orthogonal directions in the vector space of neuron outputs. Since the features are only almost-orthogonal, one feature activating looks like other features slightly activating. Tolerating this "noise" or "interference" comes at a cost. But for neural networks with highly sparse features, this cost may be outweighed by the benefit of being able to represent more features! (Crucially, sparsity greatly reduces the costs since sparse features are rarely active to interfere with each other, and non-linear activation functions create opportunities to filter out small amounts of noise.)

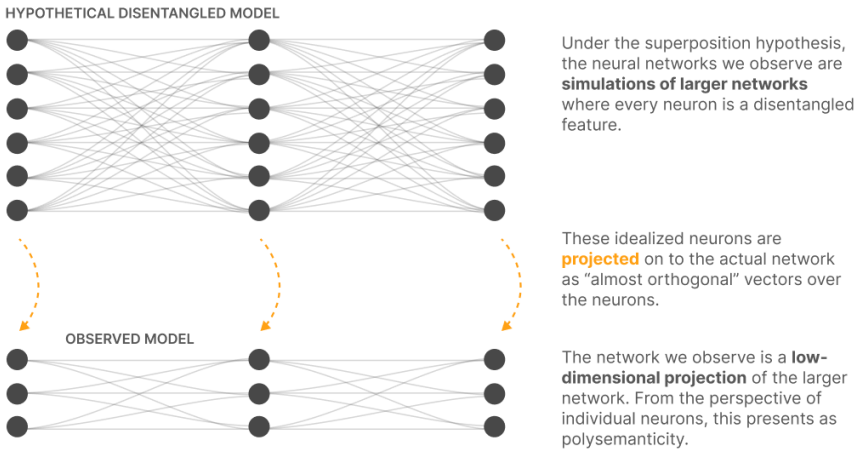


Even if only **one sparse feature** is active, using linear dot product projection on the superposition leads to **interference** which the model must tolerate or filter.



If the features aren't as sparse as a superposition is expecting, **multiple present features** can additively interfere such that there are multiple possible nonlinear reconstructions of an **activation vector**.

One way to think of this is that a small neural network may be able to noisily "simulate" a sparse larger model:



Although we've described superposition with respect to neurons, it can also occur in representations with an unprivileged basis, such as a word embedding. Superposition simply means that there are more features than dimensions.

Summary: A Hierarchy of Feature Properties

The ideas in this section might be thought of in terms of four progressively more strict properties that neural network representations might have.

- **Decomposability:** Neural network activations which are *decomposable* can be decomposed into features, the meaning of which is not dependent on the value of other features. (This property is ultimately the most important – see the role of decomposition in defeating the curse of dimensionality.)
- **Linearity:** Features correspond to directions. Each feature f_i has a corresponding representation direction W_i. The presence of multiple features $f_1, f_2 \ldots$ activating with values $x_{f_1}, x_{f_2} \ldots$ is represented by $x_{f_1} W_{f_1} + x_{f_2} W_{f_2} \ldots$.
- **Superposition vs Non-Superposition:** A linear representation exhibits superposition if $W^T W$ is not invertible. If $W^T W$ is invertible, it does not exhibit superposition.
- **Basis-Aligned:** A representation is basis aligned if all W_i are one-hot basis vectors. A representation is partially basis aligned if all W_i are sparse. This requires a privileged basis.

The first two (decomposability and linearity) are properties we hypothesize to be widespread, while the latter (non-superposition and basis-aligned) are properties we believe only sometimes occur.

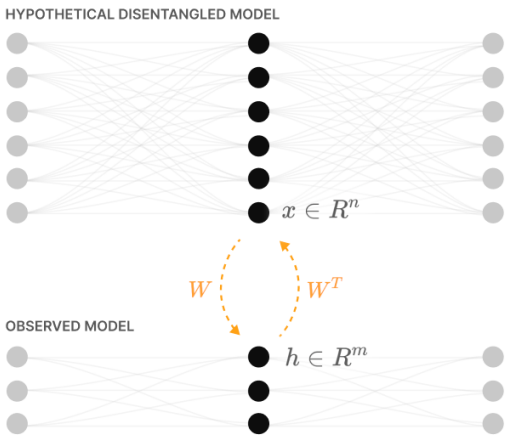
Demonstrating Superposition

If one takes the superposition hypothesis seriously, a natural first question is whether neural networks can actually noisily represent more features than they have neurons. If they can't, the superposition hypothesis may be comfortably dismissed.

The intuition from linear models would be that this isn't possible: the best a linear model can do is to store the principal components. But we'll see that adding just a slight nonlinearity can make models behave in a radically different way! This will be our first demonstration of superposition. (It will also be an object lesson in the complexity of even very simple neural networks.)

Experiment Setup

Our goal is to explore whether a neural network can project a high dimensional vector $x \in R^n$ into a lower dimensional vector $h \in R^m$ and then recover it.[5]



Our first experiments will test the extent to which the idealized activations of an imagined larger model can be **stored** and **recovered** from a lower-dimensional space.

THE FEATURE VECTOR (X)

We begin by describing the high-dimensional vector x: the activations of our idealized, disentangled larger model. We call each element x_i a "feature" because we're imagining features to be perfectly aligned with neurons in the hypothetical larger model. In a vision model, this might be a Gabor filter, a curve detector, or a floppy ear detector. In a language model, it might correspond to a token referring to a specific famous person, or a clause being a particular kind of description.

Since we don't have any ground truth for features, we need to create *synthetic data* for x which simulates any important properties we believe features have from the perspective of modeling them. We make three major assumptions:

- **Feature Sparsity:** In the natural world, many features seem to be sparse in the sense that they only rarely occur. For example, in vision, most positions in an image don't contain a horizontal edge, or a curve, or a dog head [1]. In language, most tokens don't refer to Martin Luther King or aren't part of a clause describing music [2]. This idea goes back to classical work on vision and the statistics of natural images (see e.g. Olshausen, 1997, the section "Why Sparseness?" [24]). For this reason, we will choose a sparse distribution for our features.
- **More Features Than Neurons:** There are an enormous number of potentially useful features a model might represent.[6] This imbalance between features and neurons in real models seems like it must be a central tension in neural network representations.
- **Features Vary in Importance:** Not all features are equally useful to a given task. Some can reduce the loss more than others. For an ImageNet model, where classifying different species of dogs is a central task, a floppy ear detector might be one of the most important features it can have. In contrast, another feature might only very slightly improve performance.[7]

Concretely, our synthetic data is defined as follows: The input vectors x are synthetic data intended to simulate the properties we believe the true underlying features of our task have. We consider each dimension x_i to be a "feature". Each one has an associated sparsity S_i and importance I_i. We let $x_i = 0$ with probability S_i, but is otherwise uniformly distributed between $[0, 1]$.[8] In practice, we focus on the case where all features have the same sparsity, $S_i = S$.

THE MODEL ($X \to X'$)

We will actually consider two models, which we motivate below. The first "linear model" is a well understood baseline which does not exhibit superposition. The second "ReLU output model" is a very simple model which does exhibit superposition. The two models vary only in the final activation function.

Linear Model	**ReLU Output Model**
$h = Wx$ $x' = W^T h + b$	$h = Wx$ $x' = \mathrm{ReLU}(W^T h + b)$
$x' = W^T W x + b$	$x' = \mathrm{ReLU}(W^T W x + b)$

Why these models?

The superposition hypothesis suggests that each feature in the higher-dimensional model corresponds to a direction in the lower-dimensional space. This means we can represent the down projection as a linear map $h = Wx$. Note that each column W_i corresponds to the direction in the lower-dimensional space that represents a feature x_i.

To recover the original vector, we'll use the transpose of the same matrix W^T. This has the advantage of avoiding any ambiguity regarding what direction in the lower-dimensional space really corresponds to a feature. It also seems relatively mathematically principled[9] , and empirically works.

We also add a bias. One motivation for this is that it allows the model to set features it doesn't represent to their expected value. But we'll see later that the ability to set a negative bias is important for superposition for a second set of reasons – roughly, it allows models to discard small amounts of noise.

The final step is whether to add an activation function. This turns out to be critical to whether superposition occurs. In a real neural network, when features are actually used by the model to do computation, there will be an activation function, so it seems principled to include one at the end.

THE LOSS

Our loss is weighted mean squared error weighted by the feature importances, I_i, described above:

$$L = \sum_x \sum_i I_i (x_i - x'_i)^2$$

Basic Results

Our first experiment will simply be to train a few ReLU output models with different sparsity levels and visualize the results. (We'll also train a linear model – if optimized well enough, the linear model solution does not depend on sparsity level.)

The main question is how to visualize the results. The simplest way is to visualize $W^T W$ (a features by features matrix) and b (a feature length vector). Note that features are arranged from most important to least, so the results have a fairly nice structure. Here's an example of what this type of visualization might look like, for a small model model ($n = 20;\ m = 5;$) which behaves in the "expected linear model-like" way, only representing as many features as it has dimensions: